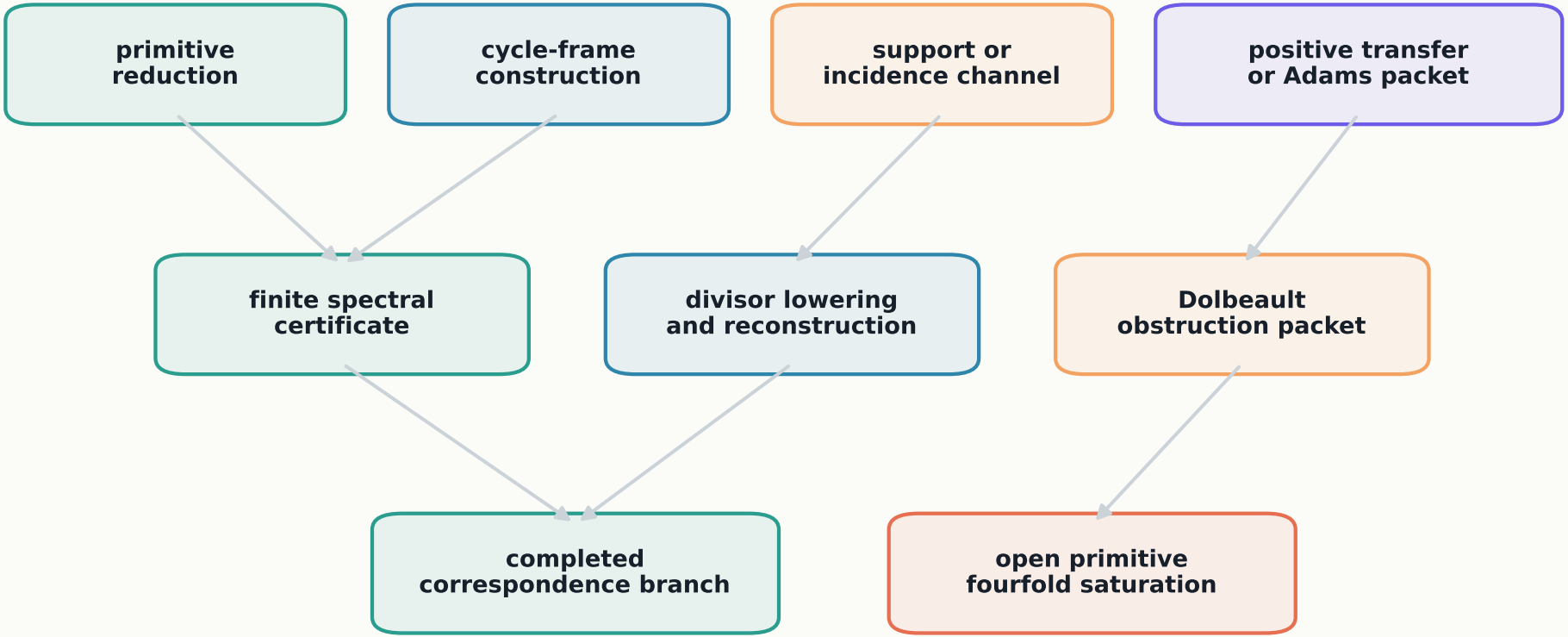Dependency graph of the Hodge program